

Practical 2

Graphical Method

Use the graphical method to solve the following Lpp

1) Maximize $z = 4x_1 + 10x_2$

 Subject to linear constraint,

$$2x_1 + x_2 \leq 50$$
$$x_1 + 3x_2 \leq 90$$
$$2x_1 + 5x_2 \leq 100$$

 And

$$x_1 \geq 0, x_2 \geq 0$$

2) Minimize $z = 3x_1 + 2x_2$

 Subject to linear constraint,

$$5x_1 + x_2 \geq 10$$
$$x_1 + x_2 \geq 6$$
$$x_1 + 4x_2 \geq 12$$

 And

$$x_1 \geq 0, x_2 \geq 0$$